



Name: _____

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Learning Objectives

- Evaluate a function at a given input value
- Compute the slope of a line given two points
- Find the composition of two functions
- Find and evaluate the inverse of a linear function

Simplify each expression completely. Show all steps and circle your final answer.

Function composition

1. Let $f(x) = 1x + -5$ and $g(x) = 3x + -2$. Find (f composed with g)(-2), i.e., $f(g(-2))$.

$$f(x) = 1x + -5, g(x) = 3x + -2, x = -2$$

Answer: _____

2. Let $f(x) = 3x + -4$ and $g(x) = 4x + 3$. Find (f composed with g)(0), i.e., $f(g(0))$.

$$f(x) = 3x + -4, g(x) = 4x + 3, x = 0$$

Answer: _____

3. Let $f(x) = 3x + -5$ and $g(x) = 5x + 2$. Find (f composed with g)(3), i.e., $f(g(3))$.

$$f(x) = 3x + -5, g(x) = 5x + 2, x = 3$$

Answer: _____

4. Let $f(x) = 1x + 0$ and $g(x) = 5x + 4$. Find (f composed with g)(-3), i.e., $f(g(-3))$.

$$f(x) = 1x + 0, g(x) = 5x + 4, x = -3$$

Answer: _____

5. Let $f(x) = 5x + -3$ and $g(x) = 1x + -2$. Find (f composed with g)(0), i.e., $f(g(0))$.

$$f(x) = 5x + -3, g(x) = 1x + -2, x = 0$$

Answer: _____

6. Let $f(x) = 1x + -4$ and $g(x) = 4x + 2$. Find (f composed with g)(0), i.e., $f(g(0))$.

$$f(x) = 1x + -4, g(x) = 4x + 2, x = 0$$

Answer: _____

Inverse of a linear function

7. Given $f(x) = 1x + -2$, find $f^{-1}(-4)$. That is, find x such that $f(x) = -4$.

$$f(x) = 1x - 2, y = -4$$

Answer: _____

8. Given $f(x) = 5x + -1$, find $f^{-1}(-11)$. That is, find x such that $f(x) = -11$.

$$f(x) = 5x - 1, y = -11$$

Answer: _____

9. Given $f(x) = 5x + -3$, find $f^{-1}(12)$. That is, find x such that $f(x) = 12$.

$$f(x) = 5x - 3, y = 12$$

Answer: _____

10. Given $f(x) = 5x + -3$, find $f^{-1}(-8)$. That is, find x such that $f(x) = -8$.

$$f(x) = 5x - 3, y = -8$$

Answer: _____

11. Given $f(x) = 4x + 1$, find $f^{-1}(1)$. That is, find x such that $f(x) = 1$.

$$f(x) = 4x + 1, y = 1$$

Answer: _____

12. Given $f(x) = 6x + 1$, find $f^{-1}(25)$. That is, find x such that $f(x) = 25$.

$$f(x) = 6x + 1, y = 25$$

Answer: _____

Evaluating a linear function

13. Given $f(x) = 7x + 8$, find $f(3)$.

$$f(x) = 7x + 8, x = 3$$

Answer: _____

14. Given $f(x) = 4x - 7$, find $f(3)$.

$$f(x) = 4x - 7, x = 3$$

Answer: _____

15. Given $f(x) = 2x + 3$, find $f(1)$.

$$f(x) = 2x + 3, \quad x = 1$$

Answer: _____

16. Given $f(x) = 5x - 5$, find $f(1)$.

$$f(x) = 5x - 5, \quad x = 1$$

Answer: _____

17. Given $f(x) = 3x + 6$, find $f(3)$.

$$f(x) = 3x + 6, \quad x = 3$$

Answer: _____

18. Given $f(x) = 2x - 4$, find $f(-2)$.

$$f(x) = 2x - 4, \quad x = -2$$

Answer: _____

19. Given $f(x) = 4x + -6$, find $f(5)$.

$$f(x) = 4x - 6, \quad x = 5$$

Answer: _____

20. Given $f(x) = 4x - 5$, find $f(5)$.

$$f(x) = 4x - 5, \quad x = 5$$

Answer: _____

21. Given $f(x) = 8x + -5$, find $f(5)$.

$$f(x) = 8x - 5, \quad x = 5$$

Answer: _____

22. Given $f(x) = 5x - 5$, find $f(0)$.

$$f(x) = 5x - 5, \quad x = 0$$

Answer: _____

23. Given $f(x) = 1x + -5$, find $f(-3)$.

$$f(x) = 1x - 5, \quad x = -3$$

Answer: _____

24. Given $f(x) = 2x - 5$, find $f(3)$.

$$f(x) = 2x - 5, \quad x = 3$$

Answer: _____

Slope between two points

25. Find the slope of the line passing through $(-4, 4)$ and $(-3, 3)$.

$$(-4, 4), (-3, 3)$$

Answer: _____

26. Find the slope of the line passing through $(-2, 8)$ and $(1, 7)$.

$$(-2, 8), (1, 7)$$

Answer: _____

27. Find the slope of the line passing through $(3, -8)$ and $(11, 0)$.

$$(3, -8), (11, 0)$$

Answer: _____

28. Find the slope of the line passing through $(2, -3)$ and $(6, -1)$.

$$(2, -3), (6, -1)$$

Answer: _____

29. Find the slope of the line passing through $(-4, 1)$ and $(3, 0)$.

$$(-4, 1), (3, 0)$$

Answer: _____

30. Find the slope of the line passing through $(0, 8)$ and $(1, 8)$.

$$(0, 8), (1, 8)$$

Answer: _____



MATH155: Functions and Graphs

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ANSWER KEY & SOLUTIONS

Topics: Slope between two points, Evaluating a linear function, Inverse of a linear function, Function composition. All answers verified by independent computation.

Solutions

Function composition

1. Let $f(x) = 1x + -5$ and $g(x) = 3x + -2$. Find (f composed with g)(-2), i.e., $f(g(-2))$.

$$f(x) = 1x + -5, \quad g(x) = 3x + -2, \quad x = -2$$

→ First compute $g(-2) = 3(-2) + -2 = -8$.

→ Then compute $f(g(-2)) = f(-8) = 1(-8) + -5 = -13$.

Answer: $(f \circ g)(-2) = f(-8) = -13$

2. Let $f(x) = 3x + -4$ and $g(x) = 4x + 3$. Find (f composed with g)(0), i.e., $f(g(0))$.

$$f(x) = 3x + -4, \quad g(x) = 4x + 3, \quad x = 0$$

→ First compute $g(0) = 4(0) + 3 = 3$.

→ Then compute $f(g(0)) = f(3) = 3(3) + -4 = 5$.

Answer: $(f \circ g)(0) = f(3) = 5$

3. Let $f(x) = 3x + -5$ and $g(x) = 5x + 2$. Find (f composed with g)(3), i.e., $f(g(3))$.

$$f(x) = 3x + -5, \quad g(x) = 5x + 2, \quad x = 3$$

→ First compute $g(3) = 5(3) + 2 = 17$.

→ Then compute $f(g(3)) = f(17) = 3(17) + -5 = 46$.

Answer: $(f \circ g)(3) = f(17) = 46$

4. Let $f(x) = 1x + 0$ and $g(x) = 5x + 4$. Find (f composed with g)(-3), i.e., $f(g(-3))$.

$$f(x) = 1x + 0, \quad g(x) = 5x + 4, \quad x = -3$$

→ First compute $g(-3) = 5(-3) + 4 = -11$.

→ Then compute $f(g(-3)) = f(-11) = 1(-11) + 0 = -11$.

Answer: $(f \circ g)(-3) = f(-11) = -11$

5. Let $f(x) = 5x + -3$ and $g(x) = 1x + -2$. Find (f composed with g)(0), i.e., $f(g(0))$.

$$f(x) = 5x + -3, \quad g(x) = 1x + -2, \quad x = 0$$

→ First compute $g(0) = 1(0) + -2 = -2$.

→ Then compute $f(g(0)) = f(-2) = 5(-2) + -3 = -13$.

Answer: $(f \circ g)(0) = f(-2) = -13$

6. Let $f(x) = 1x + -4$ and $g(x) = 4x + 2$. Find (f composed with g)(0), i.e., $f(g(0))$.

$$f(x) = 1x + -4, \quad g(x) = 4x + 2, \quad x = 0$$

→ First compute $g(0) = 4(0) + 2 = 2$.

→ Then compute $f(g(0)) = f(2) = 1(2) + -4 = -2$.

Answer: $(f \circ g)(0) = f(2) = -2$

Inverse of a linear function

7. Given $f(x) = 1x + -2$, find $f^{-1}(-4)$. That is, find x such that $f(x) = -4$.

$$f(x) = 1x - 2, y = -4$$

→ Set $f(x) = -4$: $1x + -2 = -4$.

→ Subtract -2 : $1x = -4 - -2 = -2$.

→ Divide by 1 : $x = -2 / 1 = -2$.

Answer: $f^{-1}(-4) = -2$

8. Given $f(x) = 5x + -1$, find $f^{-1}(-11)$. That is, find x such that $f(x) = -11$.

$$f(x) = 5x - 1, y = -11$$

→ Set $f(x) = -11$: $5x + -1 = -11$.

→ Subtract -1 : $5x = -11 - -1 = -10$.

→ Divide by 5 : $x = -10 / 5 = -2$.

Answer: $f^{-1}(-11) = -2$

9. Given $f(x) = 5x + -3$, find $f^{-1}(12)$. That is, find x such that $f(x) = 12$.

$$f(x) = 5x - 3, y = 12$$

→ Set $f(x) = 12$: $5x + -3 = 12$.

→ Subtract -3 : $5x = 12 - -3 = 15$.

→ Divide by 5 : $x = 15 / 5 = 3$.

Answer: $f^{-1}(12) = 3$

10. Given $f(x) = 5x + -3$, find $f^{-1}(-8)$. That is, find x such that $f(x) = -8$.

$$f(x) = 5x - 3, y = -8$$

→ Set $f(x) = -8$: $5x + -3 = -8$.

→ Subtract -3 : $5x = -8 - -3 = -5$.

→ Divide by 5 : $x = -5 / 5 = -1$.

Answer: $f^{-1}(-8) = -1$

11. Given $f(x) = 4x + 1$, find $f^{-1}(1)$. That is, find x such that $f(x) = 1$.

$$f(x) = 4x + 1, y = 1$$

→ Set $f(x) = 1$: $4x + 1 = 1$.

→ Subtract 1 : $4x = 1 - 1 = 0$.

→ Divide by 4 : $x = 0 / 4 = 0$.

Answer: $f^{-1}(1) = 0$

12. Given $f(x) = 6x + 1$, find $f^{-1}(25)$. That is, find x such that $f(x) = 25$.

$$f(x) = 6x + 1, y = 25$$

→ Set $f(x) = 25$: $6x + 1 = 25$.

→ Subtract 1: $6x = 25 - 1 = 24$.

→ Divide by 6: $x = 24 / 6 = 4$.

Answer: $f^{-1}(25) = 4$

Evaluating a linear function

13. Given $f(x) = 7x + 8$, find $f(3)$.

$$f(x) = 7x + 8, \quad x = 3$$

→ Substitute $x = 3$ into $f(x) = 7x + 8$.

$$\rightarrow f(3) = 7(3) + 8 = 21 + 8 = 29.$$

Answer: $f(3) = 29$

14. Given $f(x) = 4x - 7$, find $f(3)$.

$$f(x) = 4x - 7, \quad x = 3$$

→ Substitute $x = 3$: $f(3) = 4(3) - 7$.

$$\rightarrow = 12 - 7 = 5.$$

Answer: $f(3) = 5$

15. Given $f(x) = 2x + 3$, find $f(1)$.

$$f(x) = 2x + 3, \quad x = 1$$

→ Substitute $x = 1$ into $f(x) = 2x + 3$.

$$\rightarrow f(1) = 2(1) + 3 = 2 + 3 = 5.$$

Answer: $f(1) = 5$

16. Given $f(x) = 5x - 5$, find $f(1)$.

$$f(x) = 5x - 5, \quad x = 1$$

→ Substitute $x = 1$: $f(1) = 5(1) - 5$.

$$\rightarrow = 5 - 5 = 0.$$

Answer: $f(1) = 0$

17. Given $f(x) = 3x + 6$, find $f(3)$.

$$f(x) = 3x + 6, \quad x = 3$$

→ Substitute $x = 3$ into $f(x) = 3x + 6$.

$$\rightarrow f(3) = 3(3) + 6 = 9 + 6 = 15.$$

Answer: $f(3) = 15$

18. Given $f(x) = 2x - 4$, find $f(-2)$.

$$f(x) = 2x - 4, \quad x = -2$$

→ Substitute $x = -2$: $f(-2) = 2(-2) - 4$.

$$\rightarrow = -4 - 4 = -8.$$

Answer: $f(-2) = -8$

19. Given $f(x) = 4x + -6$, find $f(5)$.

$$f(x) = 4x - 6, \quad x = 5$$

→ Substitute $x = 5$ into $f(x) = 4x + -6$.

$$\rightarrow f(5) = 4(5) + -6 = 20 + -6 = 14.$$

Answer: $f(5) = 14$

20. Given $f(x) = 4x - 5$, find $f(5)$.

$$f(x) = 4x - 5, \quad x = 5$$

→ Substitute $x = 5$: $f(5) = 4(5) - 5$.

$$\rightarrow = 20 - 5 = 15.$$

Answer: $f(5) = 15$

21. Given $f(x) = 8x + -5$, find $f(5)$.

$$f(x) = 8x - 5, \quad x = 5$$

→ Substitute $x = 5$ into $f(x) = 8x + -5$.

$$\rightarrow f(5) = 8(5) + -5 = 40 + -5 = 35.$$

Answer: $f(5) = 35$

22. Given $f(x) = 5x - 5$, find $f(0)$.

$$f(x) = 5x - 5, \quad x = 0$$

→ Substitute $x = 0$: $f(0) = 5(0) - 5$.

$$\rightarrow = 0 - 5 = -5.$$

Answer: $f(0) = -5$

23. Given $f(x) = 1x + -5$, find $f(-3)$.

$$f(x) = 1x - 5, \quad x = -3$$

→ Substitute $x = -3$ into $f(x) = 1x + -5$.

$$\rightarrow f(-3) = 1(-3) + -5 = -3 + -5 = -8.$$

Answer: $f(-3) = -8$

24. Given $f(x) = 2x - 5$, find $f(3)$.

$$f(x) = 2x - 5, \quad x = 3$$

→ Substitute $x = 3$: $f(3) = 2(3) - 5$.

$$\rightarrow = 6 - 5 = 1.$$

Answer: $f(3) = 1$

Slope between two points

25. Find the slope of the line passing through (-4, 4) and (-3, 3).

$(-4, 4), (-3, 3)$

→ Use the slope formula: $m = (y_2 - y_1) / (x_2 - x_1)$.

→ $m = (3 - 4) / (-3 - -4) = -1 / 1 = -1$.

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{-1}{1} = -1$

26. Find the slope of the line passing through (-2, 8) and (1, 7).

$(-2, 8), (1, 7)$

→ Use the slope formula: $m = (y_2 - y_1) / (x_2 - x_1)$.

→ $m = (7 - 8) / (1 - -2) = -1 / 3 = -1/3$.

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{-1}{3} = -1/3$

27. Find the slope of the line passing through (3, -8) and (11, 0).

$(3, -8), (11, 0)$

→ Use the slope formula: $m = (y_2 - y_1) / (x_2 - x_1)$.

→ $m = (0 - -8) / (11 - 3) = 8 / 8 = 1$.

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{8}{8} = 1$

28. Find the slope of the line passing through (2, -3) and (6, -1).

$(2, -3), (6, -1)$

→ Use the slope formula: $m = (y_2 - y_1) / (x_2 - x_1)$.

→ $m = (-1 - -3) / (6 - 2) = 2 / 4 = 1/2$.

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{2}{4} = 1/2$

29. Find the slope of the line passing through (-4, 1) and (3, 0).

$(-4, 1), (3, 0)$

→ Use the slope formula: $m = (y_2 - y_1) / (x_2 - x_1)$.

→ $m = (0 - 1) / (3 - -4) = -1 / 7 = -1/7$.

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{-1}{7} = -1/7$

30. Find the slope of the line passing through (0, 8) and (1, 8).

$(0, 8), (1, 8)$

→ Use the slope formula: $m = (y_2 - y_1) / (x_2 - x_1)$.

→ $m = (8 - 8) / (1 - 0) = 0 / 1 = 0$.

Answer: $m = \frac{\Delta y}{\Delta x} = \frac{0}{1} = 0$
